The documentation and process conversion measures necessary to comply with this revision shall be completed by 16 November 1990

INCH-POUND

MIL-S-19500/455B
AMENDMENT 1
16 July 1990

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER SWITCHING
TYPES 2N5664, 2N5665, 2N5666, 2N5666S, 2N5667, AND 2N5667S
JAN, JANTX, JANTXV, AND JANS

This amendment forms a part of MIL-S-19500/455B, dated 19 January 1988, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Delete title and substitute as printed above.

1.3 and 1.4: Delete "2N5666" and "2N5667", four places, and substitute respectively "2N5666, 2N5666S" and "2N5667, 2N5667S".

PAGE 2

Table, 3.3.1, and 3.4: Delete "2N5666" and "2N5667", three places, and substitute respectively "2N5666, 2N5666S" and "2N5667, 2N5667S".

PAGE 7

4.5.2: Delete "2N5666" and "2N5667", two places, and substitute respectively "2N5666, 2N5666S" and "2N5667, 2N5667S".

Add the following paragraph:

"4.5.5 Group B accelerated life test. This test shall be conducted using one of the three options listed herein (a, b, or c) with the following conditions applying to all options: V_{CB} = 30 V dc; 96 hours minimum; T_J = +275°C.

a. T_A = +150°C, maximum.

b. P_T = 2.5 W (TO-66), P_T = 1.2 W (TO-5), T_A = +112°C or P_T adjusted to give a lot average of T_J = +275°C.

c. T_A = +25°C +3°C with P_T adjusted to give a lot average of T_J = +275°C.

PAGES 8 THROUGH 11

Table I: Delete "2N5666" and "2N5667", 18 places, and substitute respectively "2N5666, 2N5666S" and "2N5667, 2N5667S".

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
5 and 5a	5
6 and 6a	6

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PAGE 12

Table IIa, subgroup 4, intermittent operation life, conditions column: Delete "P_T = 2.5 W" and substitute "P_T = 1.2 W (TO-5), P_T = 2.5 W (TO-66)".

Table IIa, subgroup 5, accelerated steady-state operation life, conditions column: Delete and substitute "See 4.5.5".

PAGE 15

Table III: Delete "2N5666" and "2N5667", two places, and substitute respectively "2N5666, 2N5666S" and "2N5667, 2N5667S".

PAGE 16

Table III, subgroup 6, steady-state operation life: Delete "2N5664, 2N5666" and "2N5665, 2N5667", and substitute respectively "2N5664, 2N5665" and "2N5666, 2N5666S, 2N5667, 2N5667S".

Table IV: Delete "2N5666" and "2N5667", two places, and substitute respectively "2N5666, 2N5666S" and "2N5667, 2N5667S".

PAGE 17

Figure 4 title, delete and substitute: "FIGURE 4. Pulse response test circuit for types 2N5664, 2N5666, and 2N5666S."

Add note 6: "6. .02 µF capacitor may be removed during voltage adjustments."

PAGE 18

Figure 5 title, delete and substitute: "FIGURE 5. Pulse response test circuit for types 2N5665, 2N5667, and 2N5667S."

Add note 6: "6. .02 µF capacitor may be removed during voltage adjustments."

PAGES 20 AND 21

Figures 7 and 8: Delete "2N5666" and "2N5667", three places, and substitute respectively "2N5666, 2N5666S" and "2N5667, 2N5667S".

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - SH
Air Force - 11, 19, 80, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 13

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5961-1164)

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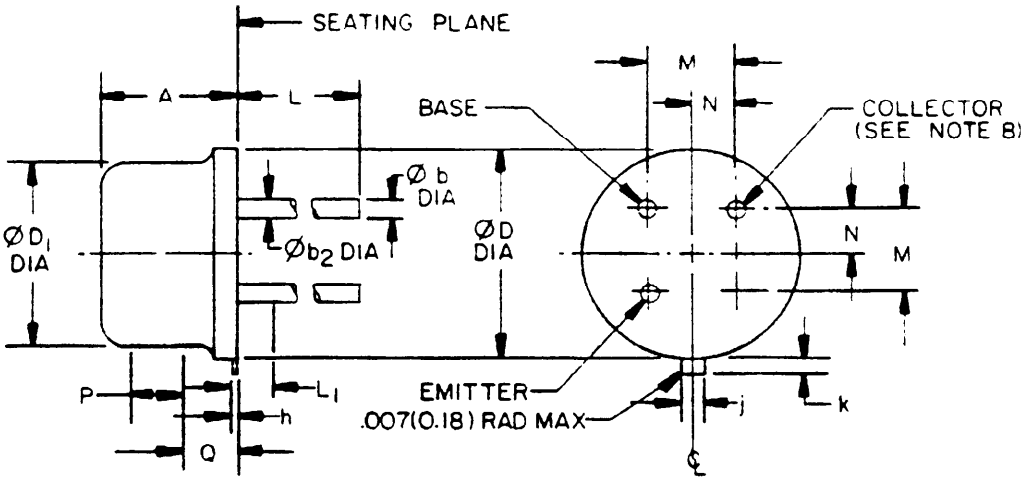


FIGURE 2. Physical dimensions of transistor types 2N5666, 2N5666S, 2N5667, and 2N5667S.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
øD	.335	.370	8.51	9.40	
$øD_1$	.305	.335	7.75	8.51	
A	.240	.260	6.10	6.60	
L	See notes 12 and 13				10
L_1		.050		1.27	11
øb	.016	.021	0.41	0.53	3,10
$øb_2$	.016	.019	0.41	0.48	4,10
P	.100		2.54		5
Q					6
k	.029	.045	0.74	1.14	9
j	.028	.034	0.71	0.86	
h	.009	.125	0.23	3.18	
M	.1414 Nom		3.592 Nom		7
N	.0707 Nom		1.796 Nom		7

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
4. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
5. Variations on dimension $øD_1$ in this zone shall not exceed .010 (0.25 mm).
6. Outline in this zone is not controlled.
7. When measured in a gauging plane .054 + .001, -.000 (1.37 +.03, -00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 (.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 3 shows the preferred measured method.
8. The collector shall be electrically connected to the case.
9. Measured from the maximum diameter of the actual device.
10. All three leads (see 3.3.1).
11. Diameter of leads in this zone is not controlled.
12. For transistor types 2N5666 and 2N5667, L is 1.500 (38.10 mm) minimum and 1.750 (44.45 mm) maximum.
13. For transistor types 2N5666S and 2N5667S, L is .500 (12.70 mm) minimum and .750 (19.05 mm) maximum.

FIGURE 2. Physical dimensions of transistor types 2N5666, 2N5666S, 2N5667, and 2N5667S - Continued.

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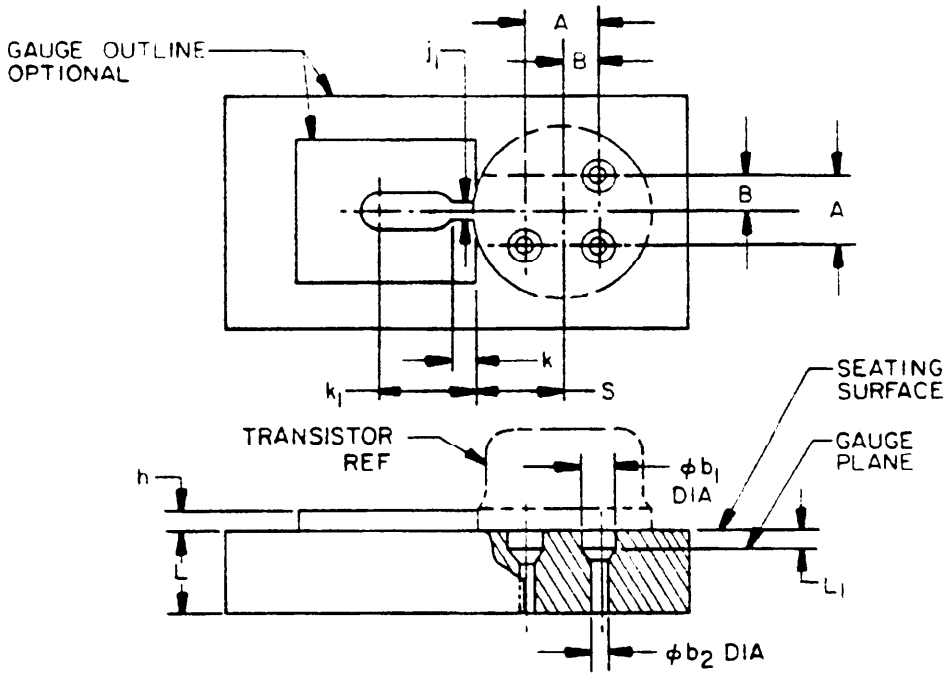


FIGURE 3. Gauge for lead and tab locations for types 2N5666, 2N5666S, 2N5667, and 2N5667S.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.1409	.1419	3.56	3.60
B	.0702	.0712	1.78	1.81
S	.182	.199	4.62	5.05
k	.009	.011	0.23	0.28
k_1	.125 Nom		3.18 Nom	
L_1	.054	.055	1.37	1.40
L	.372	.378	9.45	9.60
j_1	.0350	.0355	0.89	0.90
h	.150 Nom		3.81 Nom	
ϕb_2	.0325	.0335	0.83	0.85
ϕb_1	.0595	.0605	1.51	1.54

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The following gauging procedure shall be used: The use of a pin straightener prior to insertion in the gauge is permissible. The device being measured shall be inserted until its seating plane is .125 ±.010 (3.18 ±0.25 mm) from the seating surface of the gauge. A spacer may be used to obtain the .125 (3.18 mm) distance from the gauge seat prior to force application. A force of 6 ±.50 ounces shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed), the seating plane of the device shall be seated against the gauge.
4. The location of the tab locator, within the limits of dimension S, will be determined by the tab and flange dimension of the device being checked.

FIGURE 3. Gauge for lead and tab locations for types 2N5666, 2N5666S, 2N5667, and 2N5667S - Continued.

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